Research Meeting

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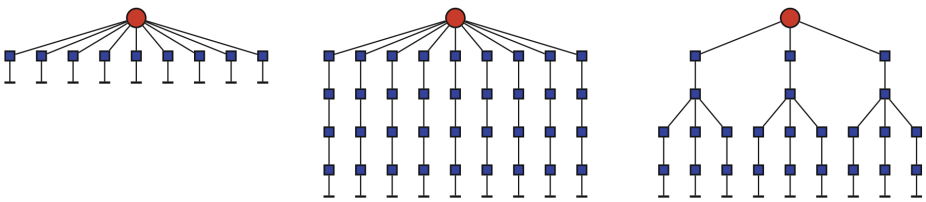
Previous Work

- Read a paper "2014_Kaut_MHSP"
- Read a chapter 1 in "Introduction to Stochastic Programming"

Today Discussion

1. **Comprehension Check**

   (a) The image below is from the Kaut paper. In (b), there are 9 operational nodes connected to strategic nodes. Am I correct in understanding that the red circles represent strategic decisions established by year and the blue boxes represent operational decisions made weekly?



(a) **(b)** **(c)**

Fig. 1 Example of scenario trees with one strategic stage and, respectively, one operational period (**a**), four operational periods (**b**), and a two-stage operational part (**c**)

   (b) What is mean 'embedded'? In section 3, Multi-horizon scenario trees. referred to the multi-horizon scenario tree structure is constructed solely out of the strategic nodes, which have the associated operation nodes 'embedded' in them.

   This is presented in Fig. 3. In this figure, tree (a) includes the same nodes as the tree in Fig. 2b, except that we interpret the nodes differently: in the new structure, the tree is constructed solely out of the strategic nodes, which have the associated operational nodes 'embedded' in them. The second tree in the figure corresponds directly to the tree from Fig. 2c. Finally, Fig. 3c demonstrates that the operational part does not need to consist of a set of deterministic scenarios (or profiles), but can itself be a multistage stochastic problem. This allows for an evaluation of the infrastructure under more complex stochastic operational conditions. In this sense, Fig. 3c provides a multi-strategic-stage equivalent of Fig. 1c.

   (c) Following up on question (b), the table below indicates the number of strategic and operational nodes for each tree structure. As can be observed, the new structure (e.g., Multi-horizon tree) significantly reduces the problem sizes. This is because when calculating the number of strategic and operational nodes, the number of operational successors is disregarded. Is this influenced by embedding the operational scenarios into the strategic scenarios?

Table 1 Numbers of strategic and operational nodes for the standard tree in Fig. 2c and the multi-horizon tree in Fig. 3b

	Standard tree	Multi-horizon tree
# strategic nodes	$\frac{1-(B_o B_s)^{P_s}}{1-B_o B_s} \approx (B_o B_s)^{P_s}$	$\frac{1-(B_s)^{P_s}}{1-B_s} \approx B_s^{P_s}$
# operational nodes	$P_o B_o \frac{1-(B_o B_s)^{P_s}}{1-B_o B_s} \approx P_o B_o (B_o B_s)^{P_s}$	$P_o B_o \frac{1-(B_s)^{P_s}}{1-B_s} \approx P_o B_o B_s^{P_s}$

P_s is the number of strategic periods, P_o the number of operational periods per strategic period. B_s and B_o are the numbers of strategic and operational successors in each strategic and operational branching, respectively. We assume that these numbers are constant throughout the tree and that $B_s > 1$

(d) How can I interpret following trees? Does it appropriate for subsequent example (i.e. installing PV panel)? Please give me a simple example to explain it.

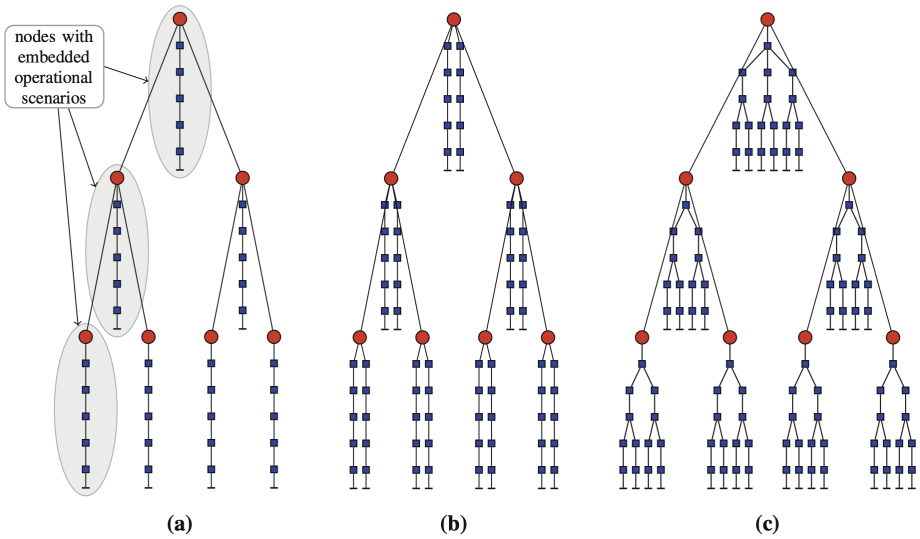


Fig. 3 Multi-horizon scenario trees with strategic uncertainty. The operational subtrees attached to the strategic nodes vary from s single operational scenario/profile **(a)**, through multiple operational scenarios **(b)**, to a stochastic multistage operational submodel **(c)**

(e) In the following object function, does x_n and y_n are the first stage and second stage of each them?

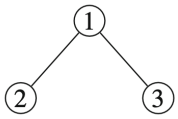
$$\min \sum_{n \in \mathcal{N}} Pr_n \Big(CI_n \, x_n + \sum_{p \in \mathcal{P}} W^p \sum_{t \in \mathcal{T}} CE_n^{p,t} \, z_n^{p,t} \Big) \tag{1}$$

$$\text{s. t.} \qquad y_n = y_{\text{Pa}(n)} + x_n \qquad \forall n \in \mathcal{N} \tag{2}$$

$$\Delta_n^{p,t} \, R_n^{p,t} \, y_n + z_n^{p,t} \geq D_n^{p,t} \qquad \forall n \in \mathcal{N}, \, p \in \mathcal{P}, \, t \in \mathcal{T} \tag{3}$$

,where x_n denotes the capacity to be installed and y_n denotes the total installed capacity.

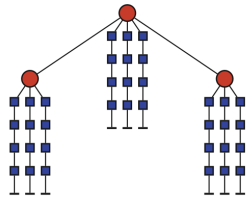
(f) Does nodes are in (a) equivalent to red circle in (c)?



(a)

n	Pa(n)	Pr_n	CI_n	SCE_n
1	–	1	700	0.05
2	1	0.5	500	0.06
3	1	0.5	600	0.04

(b)



(c)

Fig. 4 Example: the strategic tree (**a**), its parameters (**b**), and the complete multi-horizon scenario tree (**c**). Note that we show only the first four operational periods in (**c**)

(g) Following up question (f), What is SCE? and Is it applied to each node (strategic scale)? Why it is stochastic?

$$CE_n^{p,t} = SCE_n \times ICE^{p,t}$$

,where $ICE^{p,t}$ are dimensionless multipliers modelling the daily price profiles for each p.

2. **How can we find the appropriate paper for our research?**

3. **Future work**